

Assignment: Jenkins and Continuous Integration

1. What is Jenkins and Jenkins Pipeline? Explain Continuous Integration.

Jenkins

Jenkins is an open-source automation server used for **Continuous Integration (CI)** and **Continuous Delivery/Deployment (CD)** in software development. It helps developers automatically build, test, and deploy applications whenever code changes are made.

Jenkins integrates with tools like **Git, GitHub, Maven, Docker, Kubernetes, and AWS**, allowing teams to automate their development workflow. It runs jobs using plugins and supports distributed builds across multiple machines.

Key Features of Jenkins

- Open-source and free to use
 - Large plugin ecosystem
 - Automates build, test, and deployment
 - Easy integration with many DevOps tools
 - Supports distributed builds and pipelines
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Jenkins Pipeline

A **Jenkins Pipeline** is a set of automated steps that define how software is built, tested, and deployed. These steps are written as code in a file called **Jenkinsfile**.

Pipelines help teams automate the entire CI/CD workflow instead of running tasks manually.

Types of Pipelines

1. **Declarative Pipeline** – Simple and structured syntax.
2. **Scripted Pipeline** – More flexible and written in Groovy.

Example Pipeline Stages

- Code checkout from Git
- Build the application
- Run automated tests
- Deploy to staging/production

Benefits

- Automation of development process

- Faster delivery of software
 - Reduced human errors
 - Easy monitoring and debugging
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Continuous Integration (CI)

Continuous Integration is a DevOps practice where developers frequently merge their code into a shared repository. Each change triggers an automated process that builds and tests the application.

The main goal of CI is to detect errors early and ensure the code is always in a working state.

CI Workflow

1. Developer writes code
2. Code pushed to Git repository
3. Jenkins automatically triggers a build
4. Automated tests run
5. Results are shared with the team

Advantages of CI

- Early bug detection
- Improved code quality
- Faster development cycles
- Better collaboration among developers

2. Difference Between Jenkins and Other CI Tools

Feature	Jenkins	Other CI Tools
Cost	Free and open source	Many are paid or subscription-based
Plugins	1800+ plugins available	Limited integrations depending on tool
Flexibility	Highly customizable	Less customizable
Setup	Requires manual setup and maintenance	Often easier with built-in features
Hosting	Self-hosted	Many tools are cloud-based
Community	Very large community support	Smaller depending on tool

Examples of Other CI Tools

- GitHub Actions
- GitLab CI/CD
- CircleCI
- Travis CI
- Bamboo

Summary:

Jenkins is one of the most popular CI tools because of its flexibility, large plugin ecosystem, and strong community support. However, modern CI tools often provide easier cloud-based setups and built-in integrations.